

▣ A producer has the possibility of discriminating between the domestic and foreign markets for a product where the demands, respectively, are

$$Q_1 = 21 - 0.1P_1$$

$$Q_2 = 50 - 0.4P_2$$

Total cost, $TC = 2000 + 10Q$ where $Q = Q_1 + Q_2$. What price will the producer charge in order to maximize profits

(i) with discriminating between markets

Ans: Given that, in Market-1

$$Q_1 = 21 - 0.1P_1 \quad \therefore \dots \dots \dots (i)$$

$$\Rightarrow 0.1P_1 = 21 - Q_1$$

$$\Rightarrow P_1 = \frac{21}{0.1} - \frac{Q_1}{0.1}$$

$$\Rightarrow P_1 = 210 - 10Q_1 \quad \dots \dots \dots (ii)$$

$$\begin{aligned} \text{So, total revenue in market-1, } TR_1 &\equiv P_1 Q_1 = \cancel{210Q_1} - 10Q_1^2 \\ &= 210Q_1 - 10Q_1^2 \end{aligned}$$

$$\therefore MR_1 = 210 - 20Q_1$$

given that,

$$TC = 2000 + 10Q$$

$$\therefore MC = 10$$

So, profit maximization condition for ^{market} firm-1 requires

$$MC = MR_1$$

$$\text{or, } 10 = 210 - 20Q_1$$

$$\text{or, } 20Q_1 = 200$$

$$\text{or, } Q_1 = 10$$

Substituting this value into equation (1) we get

$$P_1 = 210 - 10(10) = 110$$

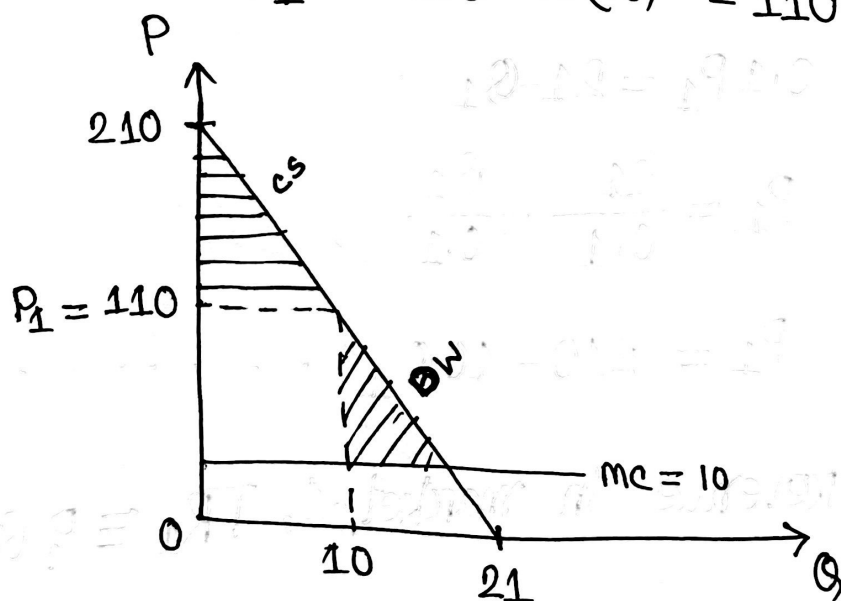


Figure (1): Profit maximizing price for firm-1 market

For Market-2

$$Q_2 = 50 - 0.4P_2$$

$$\Rightarrow 0.4P_2 = 50 - Q_2$$

$$\Rightarrow P_2 = \frac{50}{0.4} - \frac{Q_2}{0.4}$$

$$\Rightarrow P_2 = 125 - 2.5Q_2 \quad \dots \dots \dots (iii)$$

$$\begin{aligned} \text{Thus, } TR_2 &\equiv P_2 Q_2 = (125 - 2.5Q_2)Q_2 \\ &= 125Q_2 - 2.5Q_2^2 \end{aligned}$$

$$\therefore MR_2 = 125 - 5Q_2$$

So, profit maximization condition for market-2 requires

$$MC = MR_2$$

$$\Rightarrow 10 = 125 - 5Q_2$$

$$\Rightarrow 5Q_2 = 115$$

$$\therefore Q_2 = 23$$

$$\therefore P_2 = 125 - 2.5(23) = 67.5$$

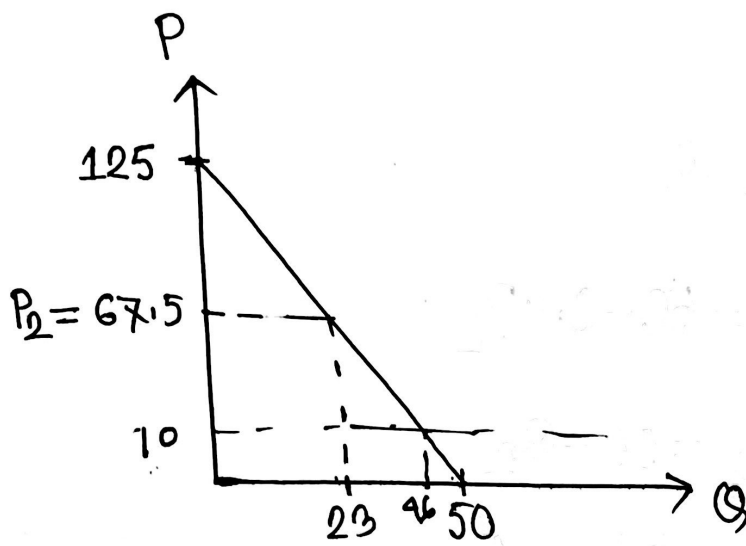


Figure (2): Profit maximizing price for market-2

Thus, producer should charge $P_1 = 110$ and $P_2 = 67.5$ in order to maximize profits with discriminating between markets, and the maximum profit is

$$\pi = TR_1 + TR_2 - TC$$

$$= (10 \times 110) + (23 \times 67.5) - 2000 - 10(10 + 23)$$

$$= 1100 + 1552.5 - 2000 - 330$$

$$= 322.5$$

(ii) Without discrimination;

If the monopoly charges single price then

$$Q_T = 21 - 0.1P + 50 - 0.4P$$

$$\Rightarrow Q_T = 71 - 0.5P$$

$$\Rightarrow 0.5P = 71 - Q_T$$

$$\therefore P = 142 - 2Q \quad \dots (iv)$$

$$\text{thus, } TR \equiv PQ = (142 - 2Q)Q = 142Q - 2Q^2$$

$$\therefore MR = 142 - 4Q$$

profit maximization condition requires that

$$MC = MR$$

$$\therefore 10 = 142 - 4Q$$

$$\Rightarrow 4Q = 132$$

$$\Rightarrow Q = 33$$

$$\therefore P = 142 - 2(33) = 76$$

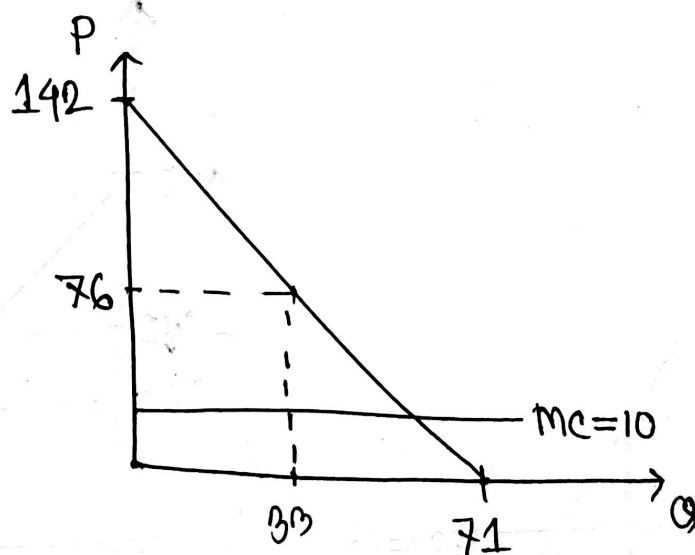


Figure (3): Profit maximizing price without discrimination

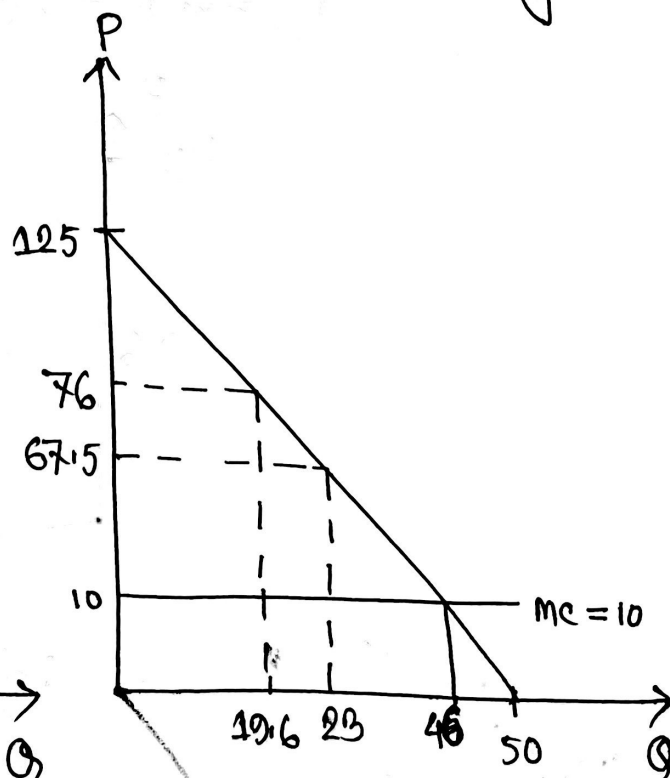
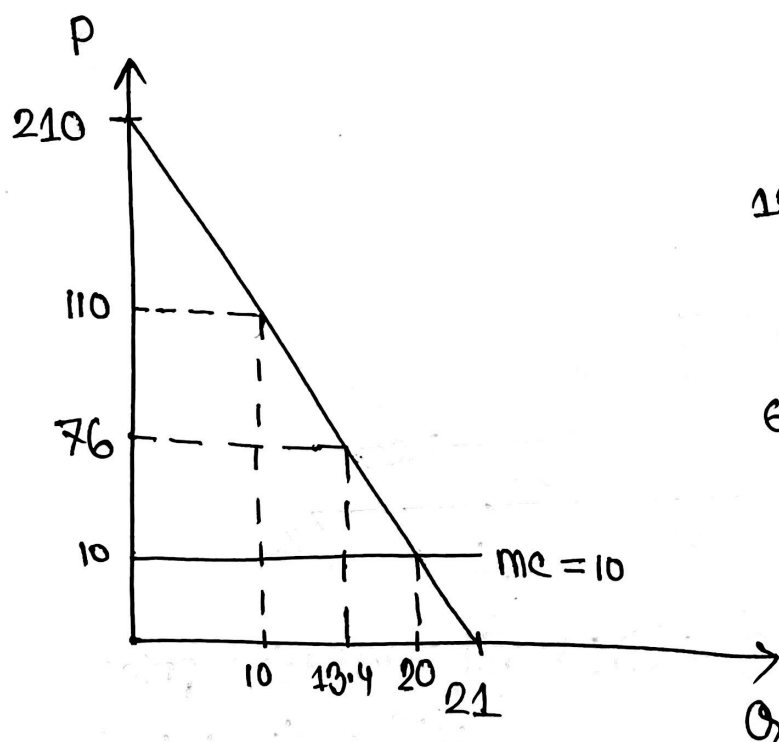
Thus, producers should charge $P=76$ in every markets

in order to maximize profits without discrimination
and the maximum profit is

$$\begin{aligned}\pi &= PQ - TC \\ &= (76 \times 33) - 2000 - 10(33) \\ &= 178\end{aligned}$$

(ii) compare the profit differential between discrimination
and nondiscrimination:

we get the profit maximizing price $P=76$ in
nondiscriminating market. Now set $P=76$ in both
market-1 and market-2 of discriminating market:



In market - 1

$$\text{initial } CS_1 = \frac{1}{2} \times 10 \times 100 = 500$$

$$\text{initial } DW_1 = \frac{1}{2} \times 10 \times 100 = 500$$

$$\text{New } CS_1 = \frac{1}{2} \times 134 \times 13.4 = 897.4$$

$$\text{New } DW_1 = \frac{1}{2} \times 66 \times 6.6 = 217.8$$

In market - 2

$$\text{initial } CS_2 = \frac{1}{2} \times 57.5 \times 23 = 661.25$$

$$\text{initial } DW_2 = \frac{1}{2} \times 57.5 \times 23 = 661.25$$

$$\text{new } CS_2 = \frac{1}{2} \times 49 \times 19.6 = 480.2$$

$$\text{new } DW_2 = \frac{1}{2} \times 66 \times 26.4 = 871.2$$

$$\text{now, new } \pi = TR_1 + TR_2 - TC$$

$$= (76 \times 13.4) + (76 \times 19.6) - 2000 - 10(13.4 + 19.6)$$

$$= 1018.4 + 1489.6 - 2000 - 330$$

$$= 2508 - 2330$$

$$= 178$$

$$\text{So, initial total surplus} \uparrow \text{ was } = CS_1 + CS_2 + \pi$$

$$= 500 + 661.25 + 322.5$$

$$= 1483.75$$

$$\text{now, } TS_2 = 897.4 + 480.2 + 178$$

$$= 1555.6$$

$$\text{initial Dead-weight loss } DW_i = 500 + 661.25$$

$$= 1161.25$$

$$\text{New Dead-weight loss } DW_n = 217.8 + 871.2$$

$$= 1089$$

Therefore, we can say that, price discrimination is profitable for producer but non discriminating price is good for welfare because total surplus is higher than price discrimination and total dead-weight loss is lower than price discrimination.